White Paper on Artificial Neural Networks and GDPR Compliance

Purpose

The purpose of this white paper is to provide information on a fictitious company's (FC) AI system, the GDPR, and to provide recommendations for compliance purposes. The paper addresses the following:

1. A brief overview of what is an artificial neural network (ANN)
2. How are ANNs used for personalization?
3. What is the impact of the GDPR on ANNs?
4. How does the GDPR impact our current business model?
5. What can we do to become more GDPR compliant?
6. Further reading

The Basics of Neural Networks

An artificial neural network (ANN) is much like its biological counterpart. Though ANNs are largely based on the mathematical relationships between nodes, those relationships mimic the signal transmissions between axons and dendrites of BNNs (Nicholson, 2023). BNNs learn through interactions and experience and are adaptable. ANNs learn through algorithms, forward/backward propagation, and adjusting weights due to errors to optimize outputs and prediction models. ANNs are largely not adaptable; they cannot rebuild pathways like BNNs (Nicholson, 2023). ANNs also only move forward through their process, which can loop, but cannot move backward or sideways, even like a BNN. Trying to mimic the complexity of a BNN with an ANN structure by increasing the number of nodes or hidden internal neurons can lead to an increase in processing time and eventually a plateau in accuracy (Gulli & Pal, 2017). Given this limitation, parallel/simultaneous processing has performance limitations (Gulli & Pal, 2017).

Neural Networks and Personalization

During the training of an ANN, individual data is fed into the system with increased weight and frequency to increase personalization of data outcomes (Schneider & Vlachos, 2020). A system like FC's cannot be trained with data from one individual at a time; similar datasets are used in data grouping (Schneider & Vlachos, 2020). Adjustments are made over time to the neural network, and as the system grows, transfer learning needs to occur. Data grouping and transfer learning together help adjust the system to seem tailored to each user (Schneider & Vlachos, 2020). Adjustments in the transfer learning process may include asking a user if they want to see more or less of a post/game/ad, and then having the algorithm adjust the weight.

There are ethical pitfalls to this process. 1) Data minimization violations. The data collected is stored and may not be needed for personalization now. However, at FC, we keep the data to help with transfer learning later, which is not data minimization. 2) Inherent bias. Data grouping can include bias, especially if data labeling is used (Wilson, Hoffman, & Morgenstern, 2019), which is why Responsible Research and Innovation (RRI) steps are needed to continuously look for ways to minimize the ethical implications that can arise from data grouping and other influences in the system (Stahl & Wright, 2018). 3) Right to Forget violations. A user could close their account or even request that their data be deleted from the system. However, the system has A) already been trained with the data from that user and B) currently, we have no means of removing a user's data from the system.

GDPR's Impact on Personalization

The GDPR impacts the personalization of AI systems on multiple levels. There are principles of the GDPR that are related to each other: purpose limitation, data minimization, and storage limitation. Transparency and consent are tied to all areas of the system.

Transparency and Consent

The GDPR mandates that users must be informed how their data is and will be used. The mandate requires this explanation to be given in language that is easy to understand for the general population (Ved, 2019). Transparency is paramount. The informed consent process must include what data is being collected, how it will be used, including tailoring content, recommendations, and services, and then an explicit opt-in process (King & Meinhardt, 2024). Transparency also requires the company to provide information on its privacy policies and on how data will be safeguarded. Further, users must have the ability to withdraw this consent at any time. To agree to use the system, the user must provide their consent.

Privacy, Purpose Limitation, and Data Minimization

Once the user has consented to the use of the system, they must have the ability to choose their privacy settings. The GDPR mandates that the default privacy settings be strong, and the user must choose to relax those settings. Data collected under the GDPR must be for a specific, explicit, and legitimate purpose, and only collected and processed for that intended purpose (King & Meinhardt, 2024; Stahl & Wright, 2018). For transparency purposes, there must be a clear description of what data will be collected, how it will be collected, and what the purpose of the collection is (King & Meinhardt, 2024; Ved, 2019). The data cannot be repurposed for any other use without additional consent from the user. Meaning, that a company must seek consent for sharing data with marketers for targeted ads if that consent was not already given by the user. Further, only the minimum amount of data to achieve the express purpose may be collected and no excessive data, outside the scope of the purpose, may be collected (King & Meinhardt, 2024; Stahl & Wright, 2018).

Storage Limitation and Bias

User data collected must not be kept for longer than is necessary for the intended purpose. AI systems must have in place data retention policies that either ensure data is deleted or is anonymized once express personalization is no longer necessary. Users must have the ability to request a copy of the data collected, how it is being used, and be able to revoke consent for its use (King & Meinhardt, 2024). Given that an AI system uses prediction models for personalization, inherent bias is an issue (Aldoseri, Al-Khalifa, & Hamouda, 2023). Though the GDPR does not expressly mention bias, they do place restrictions on profiling. Automated processes must be designed to avoid bias, be fair, and not discriminate against users (Figure Eight Federal, 2021). Users must have the ability to request human intervention to challenge a decision made by an automated process, such as blocking a social network post, or providing feedback on types of content the AI system recommended. Lastly, a user must have the ability to request their data be deleted when it is no longer needed for the intended purpose, such as an account closure (King & Meinhardt, 2024). This is often referred to as the Right to Be Forgotten.

How GDPR Impacts FC's Business Model

Currently, we collect user data from mouse clicks, site navigation, links followed, time spent on a page, location data, and more. We personalize our user experience using neural networks and algorithms to target ads and related content. Currently, we have an opt-out model regarding privacy. Additionally, the data we collect is used to train the algorithms for each user to optimize their experience. Once the data is in the algorithm, it stays. We also gather so much data that we can adequately predict which games to suggest, and our click-through rate is double our competitors meaning that although our ad-matching is effective, we are likely too broad in our data collection.

FC's business model depends on user data to complete our brand of being the most personalized social networking site available. We may be able to limit some of the data we collect, but not collecting data is not an option. Given this, we are either teetering or in direct violation of the GDPR. FC's privacy statement needs to be more robust and explicit on how we are using the data. We potentially are collecting too much data for non-descript purposes. We store data indefinitely and currently cannot delete data from the algorithm (Hill, 2020). Though we have strong security, we are using data to target ads, which may question confidentiality. We need to address these issues and point out what areas we are currently compliant with for the EU regulators.

Adaptations for GDPR Compliance

Upon review of some of the most recent policy papers on big data, ANNs, and GDPR regulations, there are adaptations we can put in place to become closer to GDPR compliance. Although there is no cure-all for GDPR compliance, some of the suggested adaptations may even set us apart from competitors. At the highest level, we can implement more accountability around the company and its use of ANNs. More granularly, we need to adjust business policies, algorithms, and the consent process. A significant compliance boost can be managed with transparency on what we are doing and a more detailed and tiered consent process. We could potentially update FC's brand message to be both the most personalized and the most privacy-conscious.

Accountability

One remedy at the highest level is to create a Data Protection Officer (DPO). The DPO would be 1) a single point of contact for regulators, 2) responsible for overseeing compliance and regulatory issues, including audits, 3) responsible for driving Responsible Research and

Innovation (RRI) efforts, and 4) handling data protection issues (Information Commissioner's Office, n.d.). Incorporating RRI as part of accountability would allow developers and architects to develop while keeping ethical implications in mind, like we already do with DevSecOps for security (Stahl & Wright, 2018).

Transparency and Bias

Our privacy policy needs to be adjusted to be clearer for the user, so they can make an informed decision on consent (King & Meinhardt, 2024). The following are key items to be clarified in simple terms. 1) How we collect and process user data. 2) The specific purpose(s) for which we are using user data. 3) What are the users consenting to regarding their data, the processing of their data, and how it relates to personalization. 4) A basic explanation of how we use ANN to create a personalized experience for the user. 5) The length of time consent is valid. 6) How a user can revoke consent. 7) How a user can request a copy of their data.

Additionally, we need to introduce additional efforts to address bias in the algorithm. We need more frequent audits to identify potential biases and actively increase the diversity in our retraining data sets for the system (IEEE, n.d.).

Purpose Limitation

We must define and communicate to users the purpose for which we collect their data. Additionally, we need to seek explicit consent to use that data for purposes other than personalization. This could be achieved with a tiered consent process, which would need to be sufficiently detailed for users to have informed consent. We need to explore controls to prevent future data from being used for purposes beyond the intended scope, for example, for individuals who did not consent to the tier that permitted future purposes (King & Meinhardt, 2024). These

controls will require adjustments to the algorithm (King & Meinhardt, 2024; Schneider & Vlachos, 2020).

Data Minimization

We need to curtail the amount of data we are collecting from users. Whatever we collect must be necessary for personalization only. We need to avoid collecting other personal information that may be irrelevant to the scope and explicitly seek consent for collecting and using personal data for future purposes (King & Meinhardt, 2024). A tiered consent process and explicit purpose limitation assist in addressing data minimization.

Storage Limitation

We need to develop data retention policies that indicate how long we are keeping user data. We need to establish a routine cycle for anonymizing, or deleting, any data that is no longer needed. Lastly, we need to establish an auditing pattern to remove outdated data from the system; outdated data can also skew our personalization system (Aldoseri, Al-Khalifa, & Hamouda, 2023). We can no longer keep data in perpetuity. A forward-thinking step, in keeping with the CCPA, we could automatically anonymize data after one year unless a user consents again to usage, like signing an annual HIPAA notice.

Quick Summary on Adaptations

Some adaptations may take longer to implement than others. However, updating our consent process, instituting an opt-in versus opt-out process for data collection, creating regular intervals for requesting updated consent, setting up an anonymization process for data over a certain age, and creating a DPO office are easier to start with and would give a big push toward GDPR compliance. Creating audit schedules and structures, retraining the algorithm, finding a way to delete data versus anonymizing it, and other initiatives are going to take longer.

References

Aldoseri, A., Al-Khalifa, K.N., and Hamouda, A.M. (2023). Re-Thinking Data Strategy and Integration for Artificial Intelligence: Concepts, Opportunities, and Challenges. Applied Sciences 13, no. 12: 7082. Retrieved from: https://doi.org/10.3390/app13127082

Figure Eight Federal. (2021). Overcome and Prevent Bias in AI. Retrieved from https://f8federal.com/overcome-and-prevent-ai-bias/

Gulli, A. & Pal, S. (2017). Deep Learning with Kera. Packt Publishing, Ltd. Retrieved from: https://ebookcentral-proquest-com.ezproxy.snhu.edu/lib/snhu-ebooks/reader.action?docID=4850536

Hill, K. (2020, January 21). The Secretive Company That Might End Privacy as We Know It. International New York Times.

Institute of Electrical and Electronics Engineers (IEEE). (n.d.). IEEE SA - IEEE Draft Standard for Algorithmic Bias Considerations. Retrieved from https://standards.ieee.org/ieee/7003/11357/

Information Commissioner's Office. (n.d.). Navigating the Accountability Framework. Retrieved from https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/accountability-and-governance/accountability-framework/navigating-the-accountability-framework/

King, J. & Meinhardt, C. (2024 February). Rethinking Privacy in the AI Era: Policy Provocations for a Data-Centric World. Human-Centered Artificial Intelligence - Stanford University. Retrieved from: https://hai.stanford.edu/white-paper-rethinking-privacy-ai-era-policy-provocations-data-centric-world

Nicholson, C.V. (2023). A Beginner's Guide to Neural Networks and Deep Learning. Pathfind. Retrieved from https://wiki.pathmind.com/neural-network

Schneider, J. & Vlachos, M. (2020, March 9). Personalization of Deep Learning. 3rd International Data Science Conference - iDSC2020: Data Science - Analytics and Applications. DOI:10.1007/978-3-658-32182-6_14

Stahl, B.C., & Wright, D. (2018). Ethics and Privacy in AI and Big Data: Implementing Responsible Research and Innovation. IEEE Computer and Reliability Societies. Retrieved from: https://ieeexplore-ieee-org.ezproxy.snhu.edu/document/8395078

Ved, A. (2019, February 28). How to Develop Artificial Intelligence That Is GDPR-Friendly. TechGDPR. Retrieved from: https://techgdpr.com/blog/develop-artificial-intelligence-ai-gdpr-friendly/

Wilson, B., Hoffman, J., & Morgenstern, J. (2019). Predictive Inequality in Object Detection. Cornell University. Retrieved from: https://arxiv.org/abs/1902.11097